BOOKNGO – AN ALL IN ONE ONLINE TRAVEL AGENCY

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

In today's digital age, online travel agencies (OTAs) provide convenient platforms for booking long-distance transportation such as buses, trains, and flights, as well as accommodations and tour packages. However, one critical aspect often overlooked by these platforms is the accessibility and guidance for local transportation—including share autos, city buses, metro routes, and bike rentals. Tourists arriving at unfamiliar destinations frequently struggle to locate nearby stops, schedules, or affordable travel options within the city, resulting in wasted time and diminished travel experiences.

BookNGo aims to address this gap by integrating local transport navigation and discovery into a unified travel platform. The system not only allows users to book intercity travel and stays but also helps them seamlessly continue their journey within the city by identifying nearby local transport options, routes, and estimated travel times. By enhancing last-mile connectivity and saving valuable time during holidays, BookNGo promotes smarter, more efficient, and traveler-friendly mobility solutions.

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CHAPTER 1:INTRODUCTION

1.1 Overview of BookNGo Project

BookNGo is a web-based travel assistant designed to simplify and enhance both long-distance and local travel experiences. Unlike traditional travel platforms that focus only on booking flights, trains, or buses, BookNGo integrates local transportation options such as share autos, city buses, and bike rentals. This allows users to plan their entire journey—from intercity travel to local commuting—within a single platform. By bridging the gap between online bookings and on-ground navigation, BookNGo ensures greater convenience, time efficiency, and a seamless travel experience for every user.

1.2 Objectives of the Project

The main objectives of the BookNGo project are as follows:

- To integrate intercity and local transport booking within a single unified platform.
- To guide users to nearby share auto stands, bus stops, and bike rental points through location-based services.
- To optimize travel time for tourists and commuters during their trips and holidays.
- To enhance local travel discovery by integrating interactive maps and route visualization.

1.3 Scope and Importance

The project focuses on improving last-mile connectivity, particularly for tourists who are unfamiliar with local routes and transport systems. BookNGo ensures seamless mobility by integrating nearby public and shared transport options, helping users navigate easily within cities. Additionally, it promotes sustainable travel by encouraging the use of eco-friendly shared and public transportation, reducing congestion and enhancing overall travel efficiency.

1.4 System Architecture Overview

The proposed BookNGo architecture follows a client–server model, designed to ensure scalability, interactivity, and real-time data access. The frontend is developed using HTML, CSS, and AngularJS, providing a dynamic and responsive user interface. The backend server created using NodeJS , Express and MongoDB manages data communication and integrates with external APIs, while the OpenStreet Map API is utilized for real-time location display and navigation assistance.

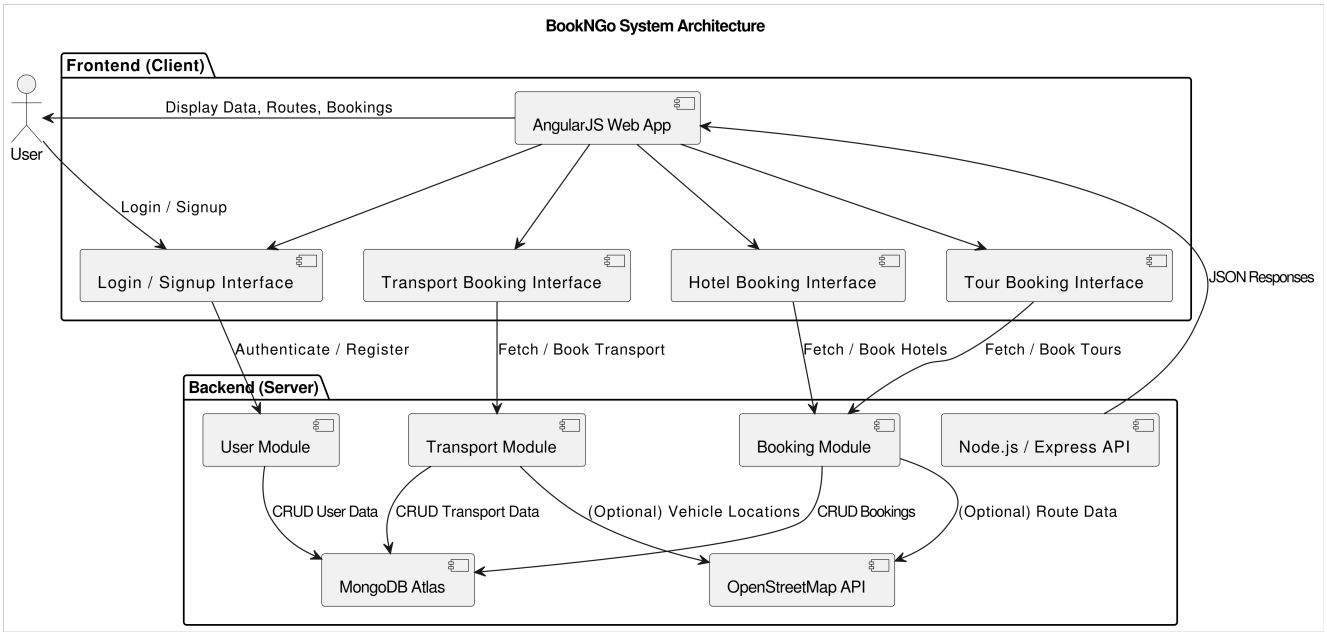


Figure 1.1: System Architecture Diagram

CHAPTER 2: EXISTING SYSTEM ANALYSIS

2.1 General Analysis of Current Web Services

Existing Online Travel Agencies (OTAs) such as MakeMyTrip, Goibibo, and RedBus primarily focus on enabling users to book tickets for long-distance travel and reserve hotel accommodations. However, these platforms lack support for intra-city mobility, leaving travelers without guidance on local transportation options like share autos, city buses, or bike rentals once they arrive at their destination. This limitation creates a gap in providing a complete end-to-end travel experience.

2.2 Limitations of Current Systems

The major limitation of current online travel platforms is the lack of integrated local transport information. Travelers often need to depend on multiple external applications or manual searches to find nearby bus stops, share auto stands, or rental services. This fragmented approach leads to inconvenience, confusion, and delays, especially for tourists unfamiliar with the area.

2.3 User Experience Challenges

Tourists often face difficulty finding nearby public transport or rental options due to the lack of integrated maps or guidance. Existing apps focus on long-distance travel and rarely provide real-time information for last-mile connectivity. Users struggle with planning multi-modal journeys, understanding transport availability, and navigating unfamiliar routes. Language barriers and scattered information from multiple sources further reduce usability and increase travel stress.

CHAPTER 3:PROPOSED SYSTEM & NOVELTY

3.1 Introduction to Proposed System

BookNGo integrates local transport information and booking options within a single interface, enabling tourists to locate nearby transport, plan multi-modal journeys, and navigate seamlessly. Real-time updates and an interactive map reduce travel uncertainty and enhance user convenience.

3.2 New Features

BookNGo offers the following features for enhanced last-mile connectivity:

- Share Auto Stops Integration: Displays nearby share-auto pickup and drop-off points.
- Local Bus Stop Mapping: Shows nearby bus stops with route information.
- Bike Rental Availability: Indicates available rental bikes in real time.
- Route Visualization with OpenStreet Map API: Provides interactive maps for seamless navigation and multi-modal route planning.

3.3 System Workflow

The BookNGo system begins with user login or registration. After authentication, the user selects the source and destination locations. The system then fetches available transport options, including share-autos, local buses, and bike rentals, and visualizes possible routes using Google Maps API. Finally, the user can confirm their chosen trip, completing the workflow from planning to booking in a seamless, interactive manner.

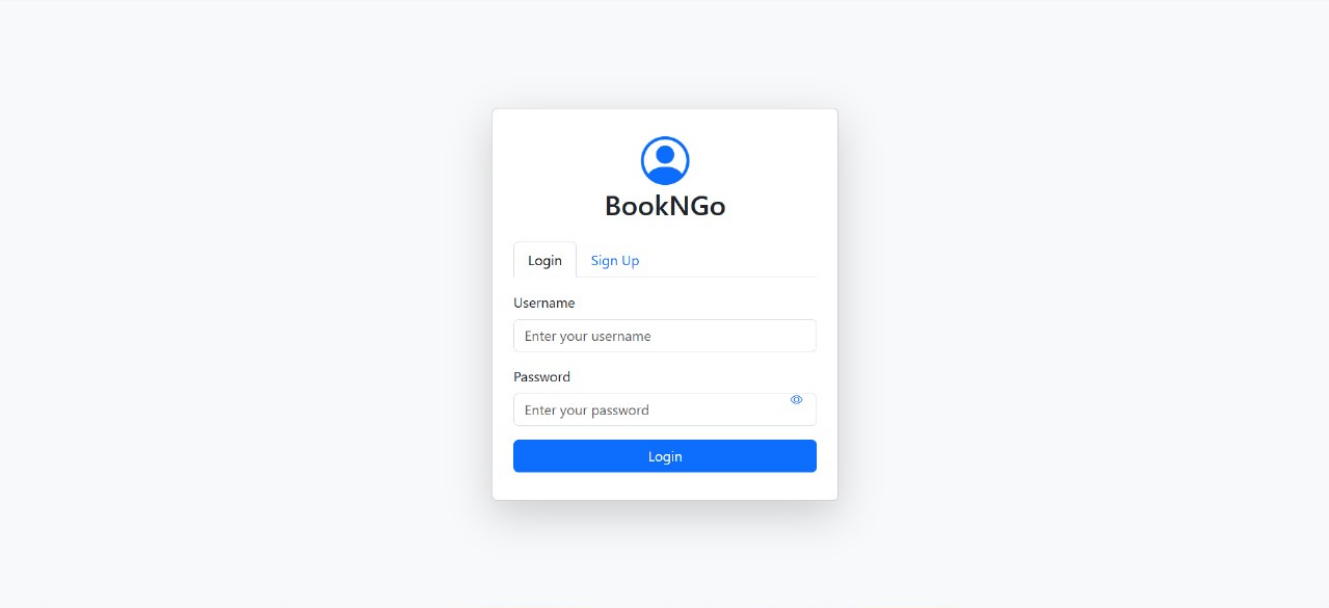


Figure 3.1: Login Page

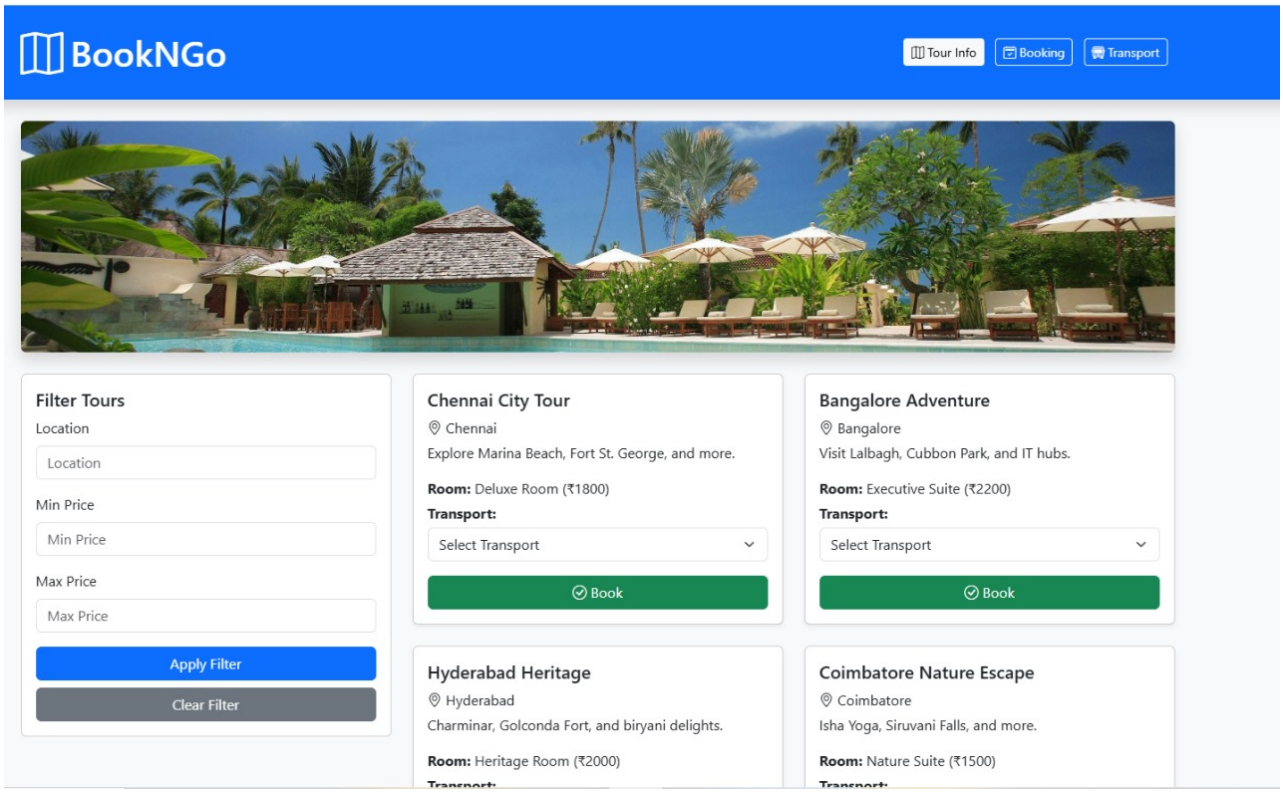


Figure 3.2: Tour Booking Page

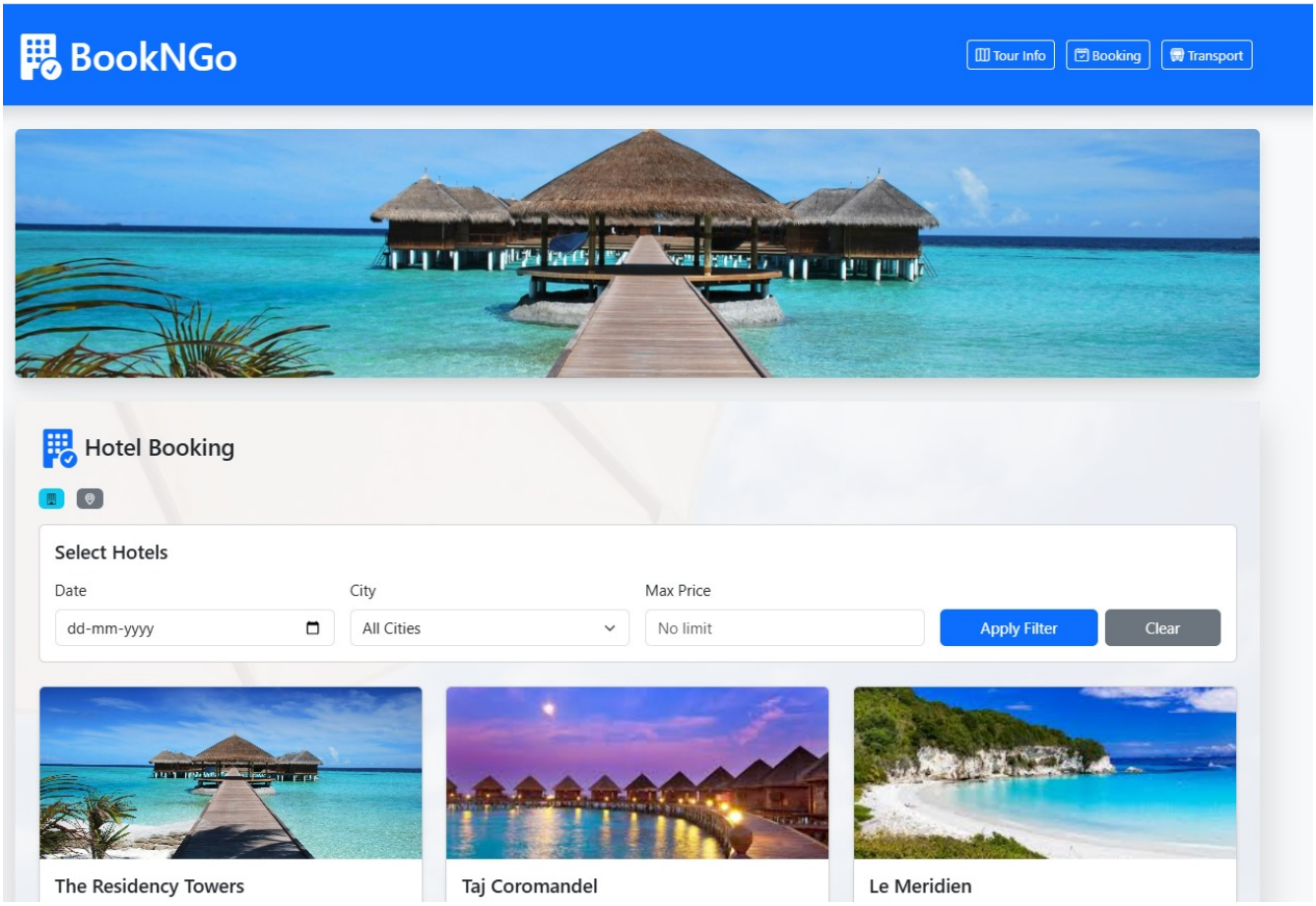


Figure 3.3: Hotel Booking Page

3.4 Improvements Over Existing Systems

The BookNGo system significantly reduces last-mile confusion and enhances travel planning efficiency by providing real-time transport options, route visualization, and seamless booking in a single interface.

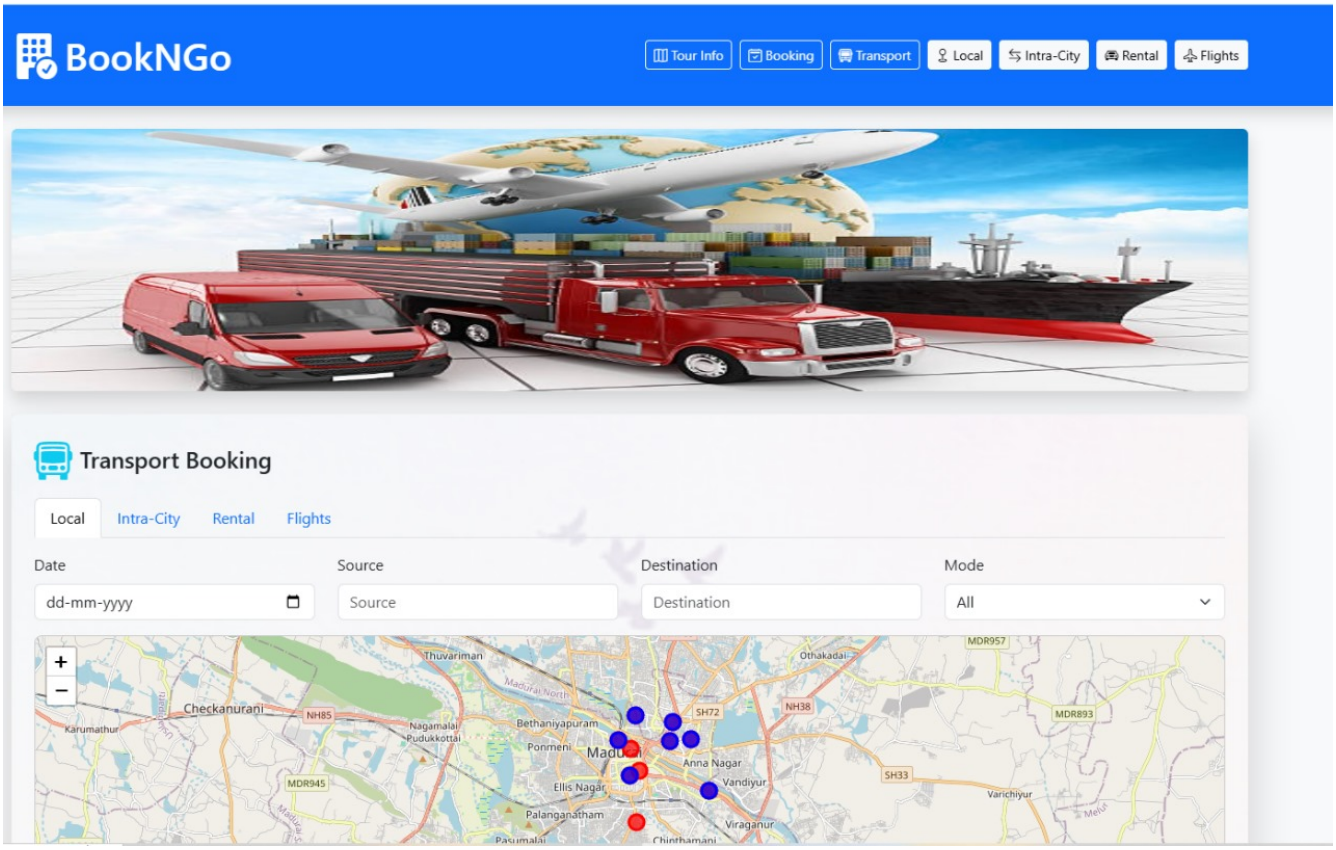


Figure 3.4: BookNGo Transport Search Page

CHAPTER 4: COMPARATIVE EVALUATION

4.1 Comparison of Existing OTAs

Platform	Features	Local Transport Support	Ease of Use
MakeMyTrip	Flight, Train, Bus, Hotel	No	High
Goibibo	Flight, Train, Bus, Hotel	No	High
RedBus	Bus Booking	Limited (Bus Routes)	High
Yatra	Flight, Train, Bus, Hotel	No	High
BookNGo (Proposed)	Flight, Train, Bus, Hotel + Local Transport	Yes	Very High

Table 4.1: Comparative Evaluation of Travel Platforms

Most existing online travel agencies (OTAs) focus on long-distance travel and accommodations, offering little or no support for intra-city transport. Platforms like RedBus provide bus routes but still lack integration with last-mile options such as bike rentals or share-autos. BookNGo, in contrast, bridges this gap by integrating local transport, real-time vehicle availability, and interactive route visualization in a single interface. This comprehensive approach enhances ease of use, reduces travel uncertainty, and ensures seamless planning for both tourists and local commuters.

CHAPTER 5: FINDINGS & RECOMMENDATIONS

The analysis reveals a significant gap in integrating local transport into online booking systems, which leaves tourists and first-time city visitors with fragmented, confusing, and inefficient intra-city travel options.

Findings:

- Most existing OTAs primarily focus on long-distance travel, offering flights, trains, and hotels, but provide minimal or no support for local transport.
- Users struggle to plan multi-modal journeys due to dispersed information, lack of real-time updates, and absence of guidance for last-mile connectivity.

Recommendations:

- Map-Based Navigation: Implement interactive maps to display nearby transport options, available routes, and estimated travel times.
- Local Data Access: Provide real-time updates on buses, share-autos, bike rentals, and other local mobility options to reduce uncertainty.
- Seamless Booking: Enable end-to-end trip planning, including local transport, within a single interface to improve convenience and user experience.

CHAPTER 6: LIMITATIONS & FUTURE WORK

The current BookNGo prototype relies on static data for bus stops, share-auto locations, and bike rentals. This limits the accuracy of route suggestions and real-time travel planning.

Limitations:

- Lack of dynamic route updates from official transport APIs.
- No live GPS tracking of vehicles, which reduces real-time reliability.
- Limited integration with other local mobility services beyond buses, autos, and bikes.

Future Work:

- Integrate dynamic route fetching and schedules using transport authority APIs.
- Implement real-time GPS tracking for buses, share-autos, and rental bikes to enhance accuracy.
- Expand multi-modal transport support to include ride-sharing, metro, and on-demand mobility services.
- Introduce predictive suggestions based on traffic conditions and historical travel patterns.

These improvements will make BookNGo a fully dynamic and reliable last-mile travel solution, further enhancing usability and user satisfaction.

CHAPTER 7: CONCLUSION

BookNGo redefines digital travel assistance by integrating long-distance bookings with local transport guidance, providing a unified platform for end-to-end travel planning. The system demonstrates how technology can enhance the traveler experience by simplifying route selection, offering real-time information, and supporting multi-modal transport options. By addressing gaps in existing OTAs, BookNGo reduces confusion, improves efficiency, and delivers a seamless, user-friendly solution for intra-city mobility.